PROJECT CASE STUDY
WEB PLATFORM · 2026

BagMech Bengaluru

A doorstep luggage-repair brand, a booking website that finds the customer's front door, and a backend that alerts the owner the moment a request comes in.



CLIENT	INDUSTRY	BUILD WINDOW	BUILT BY
BagMech Bengaluru	Doorstep repair services	Jul – Aug 2026	CraftLanee

00 · OVERVIEW

At a glance

What we built, who it was for, and where to find each part in this document.

01 The client & the brief 03

02 Design language 04

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PROJECT TICKET

BagMech Bengaluru

Service	Doorstep luggage repair
City	Bengaluru, Karnataka
Scope	Brand site · Booking · Backend
Frontend	HTML · CSS · Vanilla JS
Backend	Python · FastAPI
Database	PostgreSQL (Supabase)
Alerts	Email + WhatsApp
Hosting	Render (cloud PaaS)
Status	• Live

CL · BM · 2026

16
Bengaluru zones mapped with ETAs

6
Repair services, bookable online

2
Instant alert channels to the owner

1
Deployable app — site and API together

A repair service that comes to you — it needed a website that works the same way

BagMech Bengaluru repairs suitcases and travel bags at the customer's doorstep: wheels, trolley handles, zips, zip runners, shoulder straps and TSA locks. The technicians are mobile, so the site's real job is getting an accurate location and a clear repair request to them fast.

The challenge

- Customers usually find a broken bag the night before a trip, so they need an answer quickly.
- Phone bookings lose details: vague addresses, wrong area names, unclear repairs.
- Travel times differ across the city, and customers dislike surprise waits.
- Small repair shops rarely look trustworthy online, so the brand needed a premium feel.
- The owner is out on jobs and has to see new bookings on the phone right away.

What we delivered

- A premium, dark-and-gold brand website built around a "service ticket" idea.
- A booking form that **requires live GPS location**, so every request comes with coordinates.
- A coverage map of 16 zones with **upfront ETAs** of 25 or 45 minutes.
- Instant **email and WhatsApp alerts** to the owner, with a Google Maps link to the customer.
- WhatsApp click-to-chat that pre-fills the customer's booking details.

Design principles we worked to

01

Honest by default

ETAs are shown before booking, and far zones are flagged in orange so no customer is surprised at the door.

02

Location first

GPS is step one. The technician gets exact coordinates, so nobody has to guess the address.

03

Premium, not corporate

Editorial type, a gold accent and a tactile luggage-tag motif make a local service feel established.

Travel-gear luxury: midnight, brass and paper tags

The look takes its cues from premium luggage: a near-black base, brushed-gold hardware and cream paper tags. Teal marks live or "on the way" states, and rust marks longer ETAs.

Midnight	Surface	Brass Gold	Light Gold	Live Teal	Far-zone Rust
#07080B	#131824	#C9A25C	#EFCB86	#5FC9A8	#E2694A

Aa

FRAUNCES

Headlines. An editorial serif that gives a boutique feel.

Aa

INTER

Body and UI. Neutral and easy to read on small phones.

25m

IBM PLEX MONO

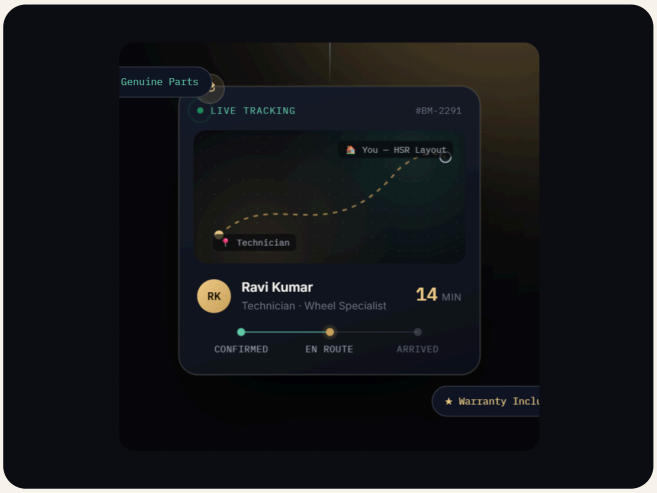
ETAs, labels and ticket codes. It reads like a printed baggage slip.

Signature element: the flipping service ticket

The hero features a cream luggage tag that swings gently and tilts in 3D with the mouse. Tapping it flips it to the back, an illustrated live-tracking card showing a technician's route to the doorstep. It explains the service in one image: *book it, and someone comes to you.*



FRONT Service ticket with a "Certified" seal, zone, ETA and status.



BACK Illustrated tracking view: route, technician and progress steps.

Film-grain texture · Aurora gradients · Scroll reveals · Magnetic buttons · Custom cursor · Synthesised UI sounds (mutable) · Reduced-motion support

First screen: the whole offer in five seconds

The headline states the promise, "We'll come fix it, not the other way round." Two buttons lead to a booking form and to WhatsApp, and three counters answer the usual first questions: how fast, where, and whether the parts are genuine.

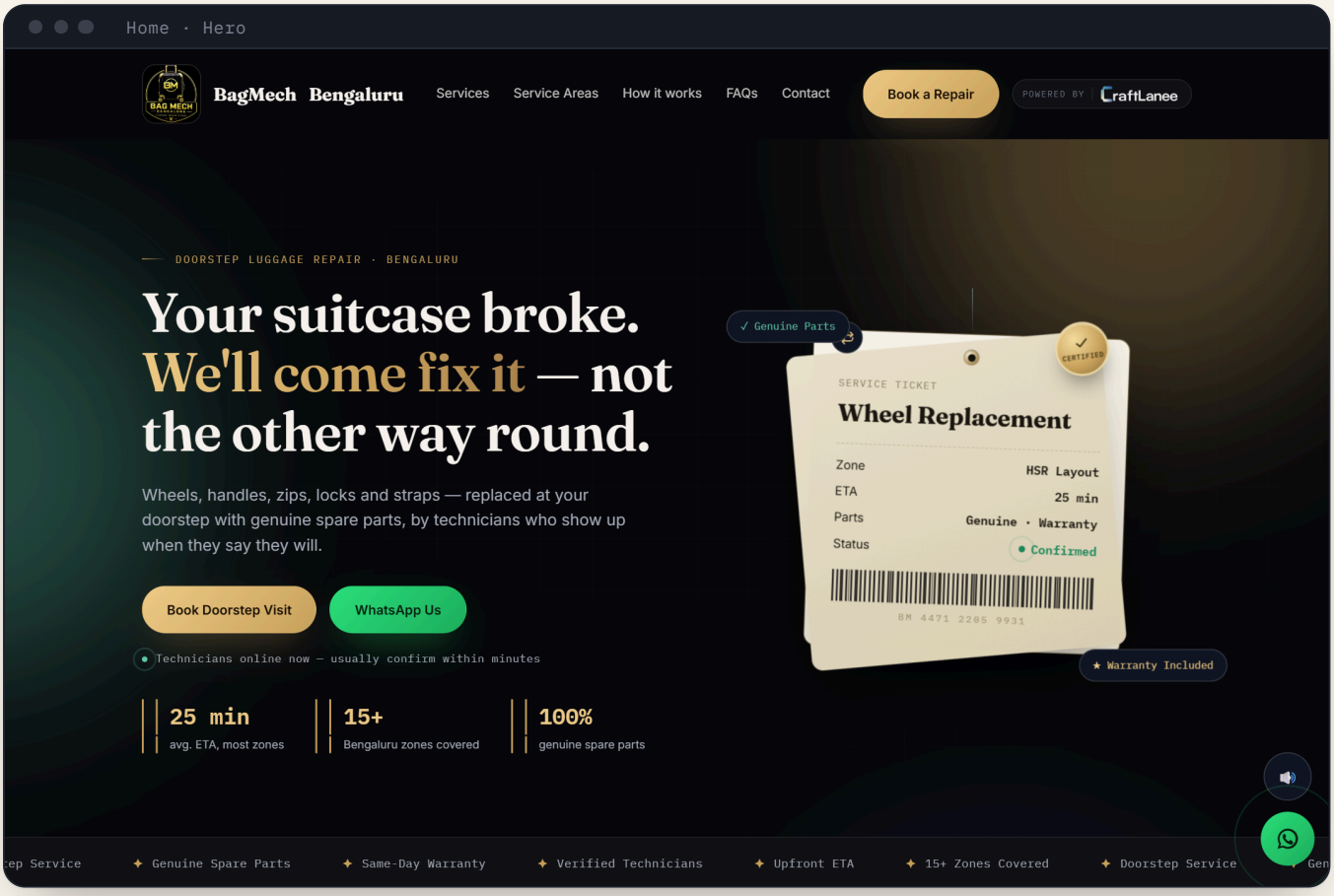


FIG 3.1 Desktop hero: headline, dual call-to-action, live "technicians online" indicator, animated stats and the service-ticket card.

A

Sticky smart header

Turns frosted-glass on scroll, with a "Book a Repair" button that stays in view.

B

Trust marquee

A scrolling strip below the hero: doorstep service, genuine parts, warranty, verified technicians.

C

Floating WhatsApp

A pulsing WhatsApp button stays one tap away on every screen.

Services and the four-step visit

Six numbered service cards, each with its own icon and a mouse-following spotlight. Below them, a four-step process strip tells the customer what happens after they book.

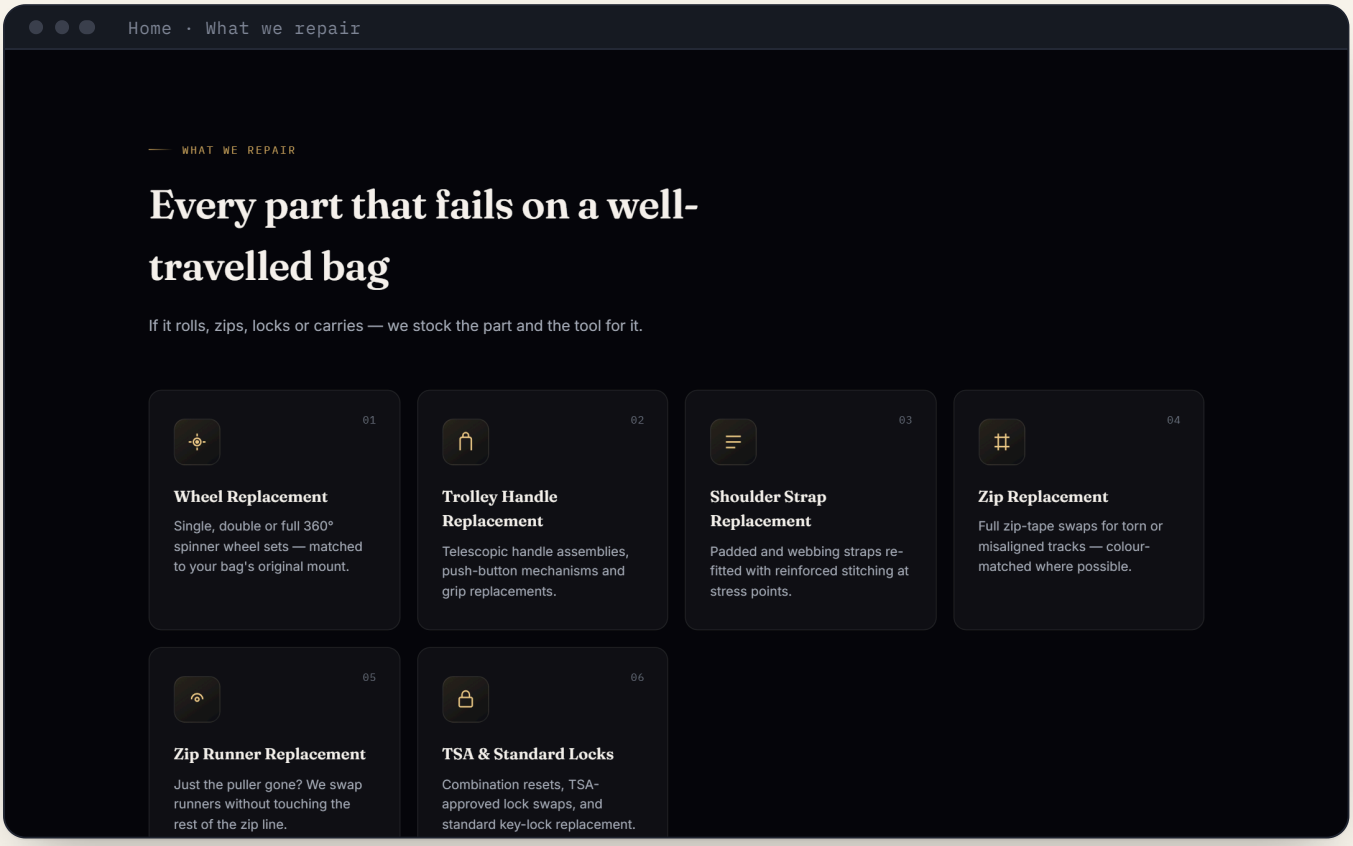


FIG 3.2 Services: wheels, trolley handles, shoulder straps, zips, zip runners, and TSA and standard locks.

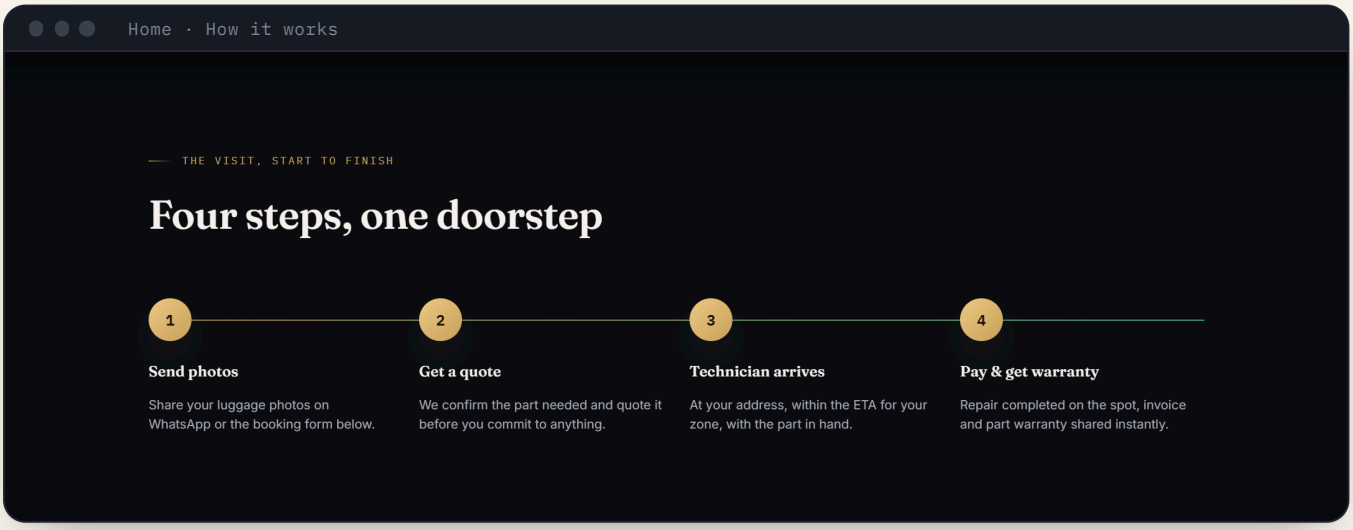


FIG 3.3 Send photos, get a quote, the technician arrives, then pay and receive a warranty. A progress line fills in as the section scrolls into view.

Coverage with honest ETAs

Every zone the team covers appears as a card with its average arrival time. The four zones farthest from the Bommanahalli / Silk Board base are marked in rust and quote 45 minutes, not the usual 25. The list is served from the database, so the owner can add a zone without touching the website code.

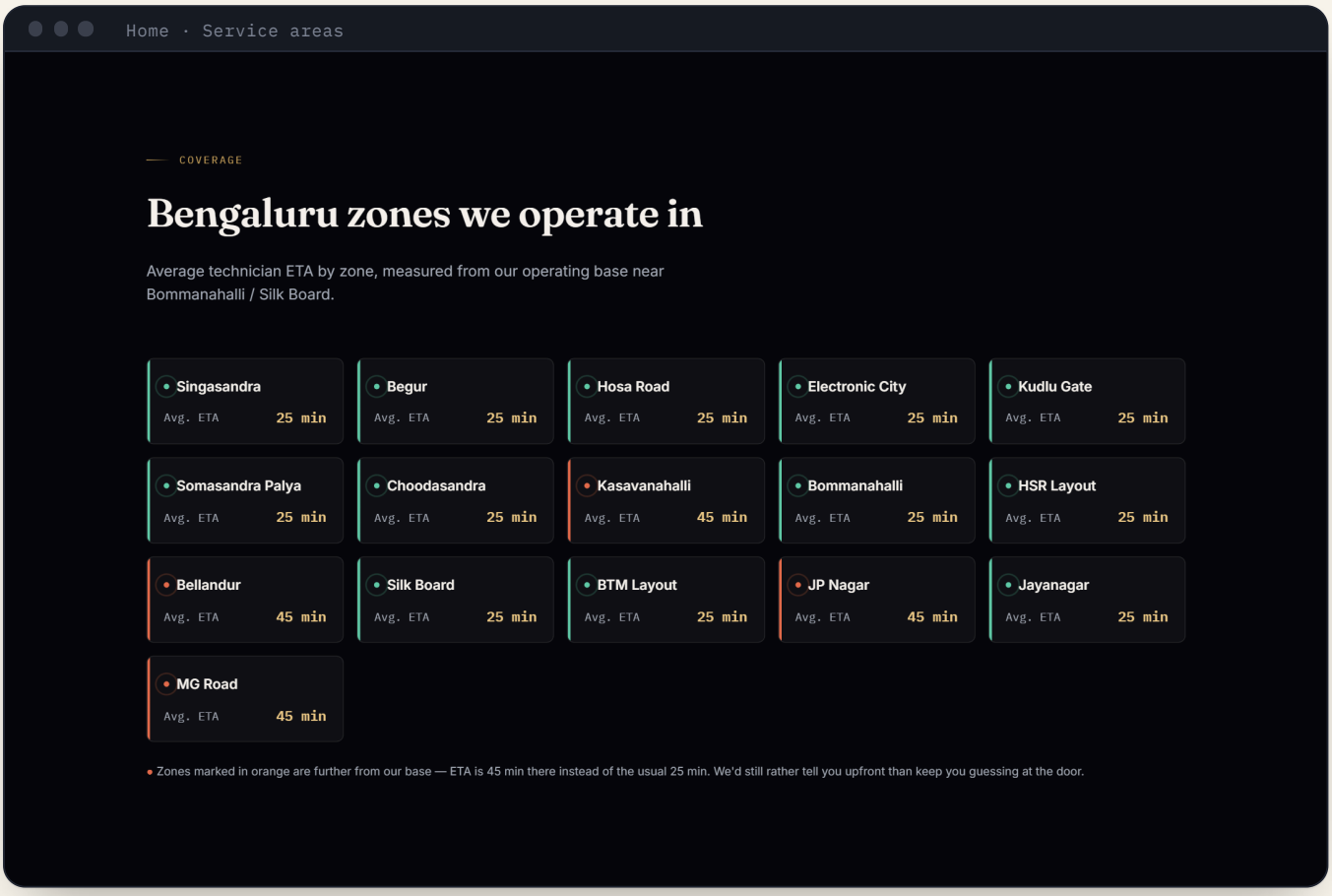


FIG 3.4 16 zones: teal cards are about 25 minutes, rust cards about 45 minutes (Kasavanahalli, Bellandur, JP Nagar, MG Road).

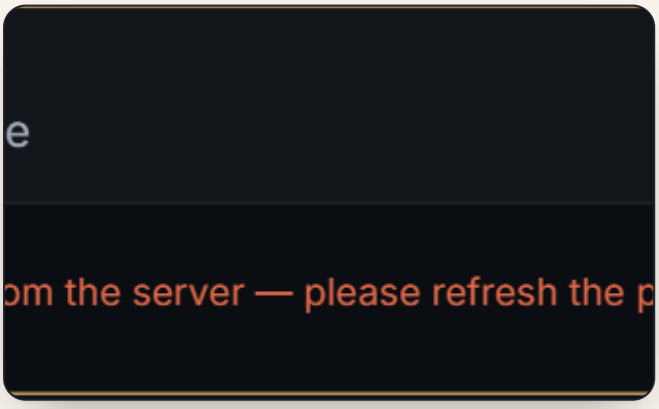


FIG 3.5 The same zones reused as a picker in the booking form.

Built to change easily

Zones live in a `zones` table with a name, an ETA in minutes and a "far" flag. The website loads them from `/api/zones` and falls back to a built-in list if the API is unreachable, so the page never shows an empty coverage section.

The booking form: location first, then details

The main feature of the project. Customers share their live location first, then fill in name, phone, address, zone and one or more repair types. A booking cannot be sent until a location has been shared.

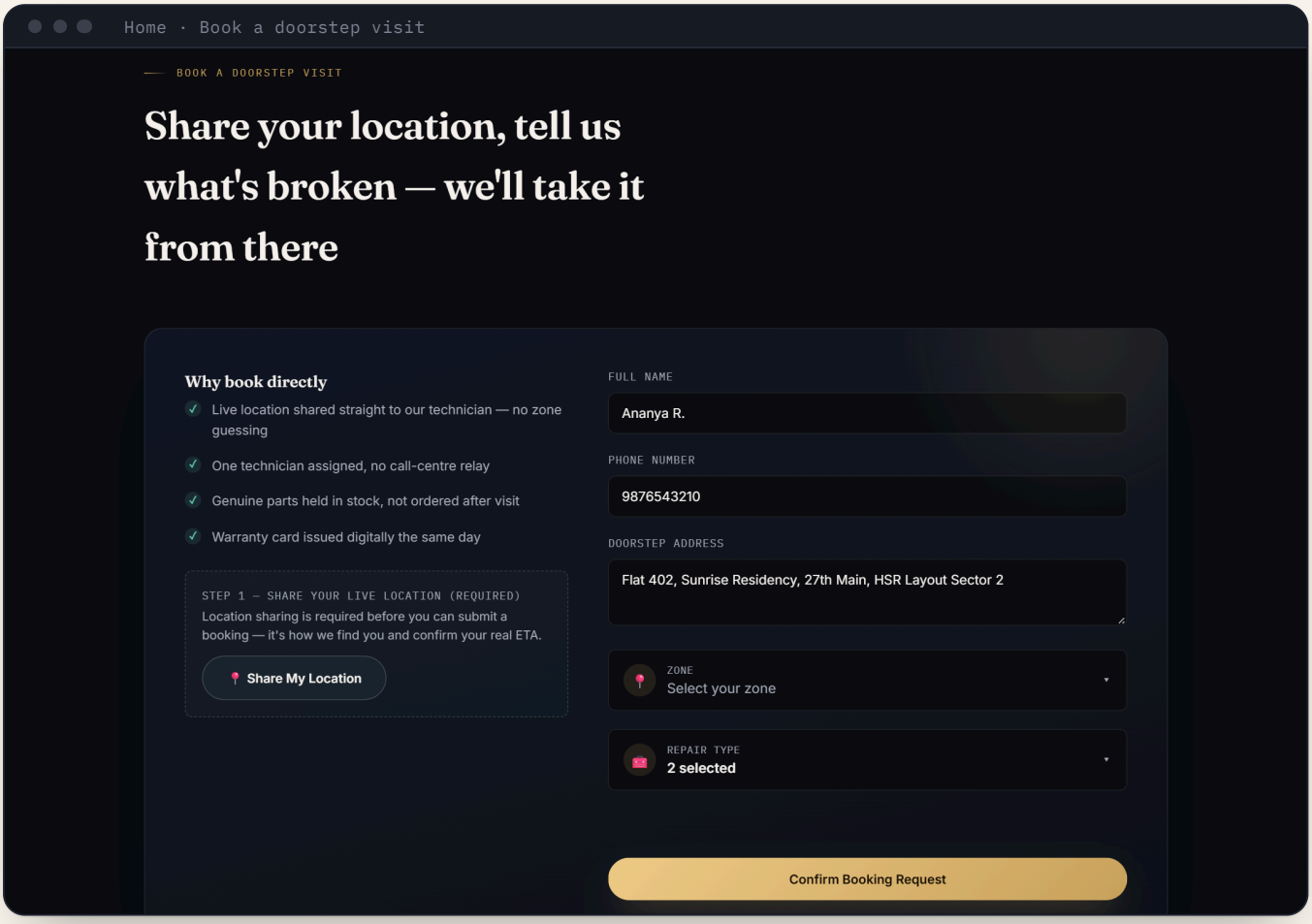


FIG 3.6 Booking panel with sample data: "Why book directly" points, the GPS step, and fold-out zone and repair-type selectors.

GPS required

Uses the browser's location permission. Clear messages cover every case: granted, denied or unsupported.

Validated twice

Indian mobile format (10 digits starting 6–9) is checked in the browser. Lengths and required fields are checked again on the server.

Multi-select repairs

Pick several repairs at once. Each shows as a chip you can remove with one tap.

Reviews, FAQs, contact and footer

The lower half of the page handles doubts: customer reviews, direct answers to common questions, the business's contact details and a full footer.

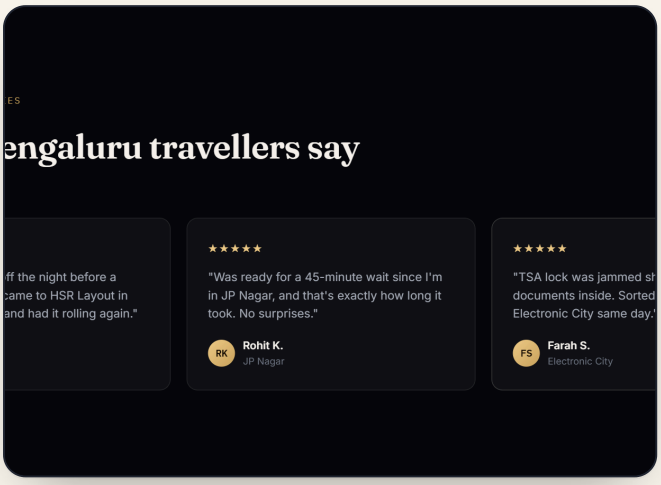


FIG 3.7 Customer stories, tagged by neighbourhood.

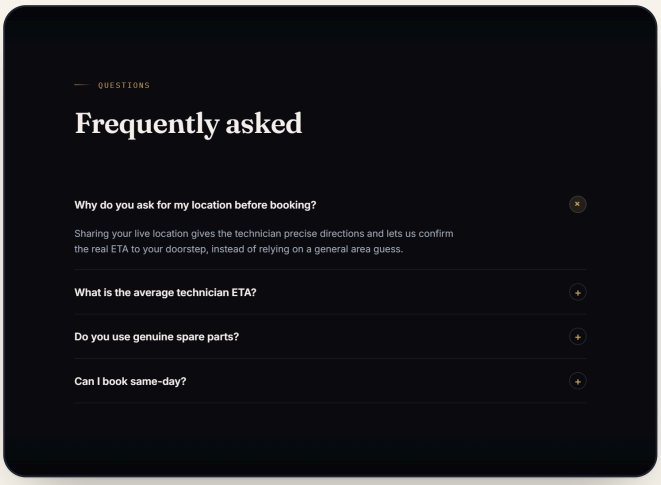


FIG 3.8 Smooth-opening FAQ answering the location, ETA, parts and same-day questions.

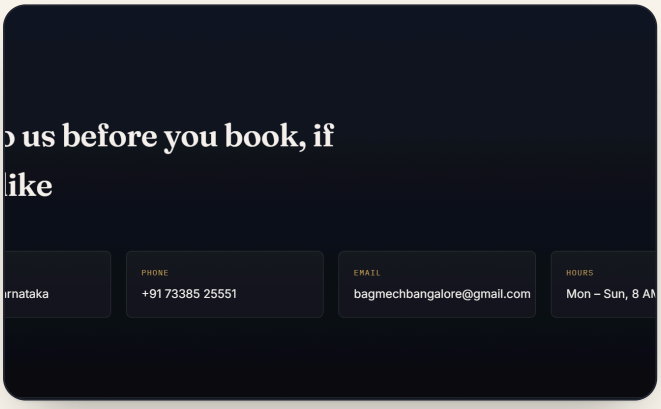


FIG 3.9 Contact band: phone, email, city and opening hours.

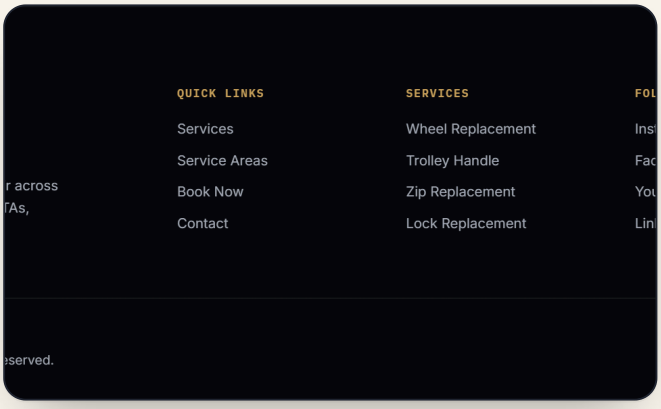
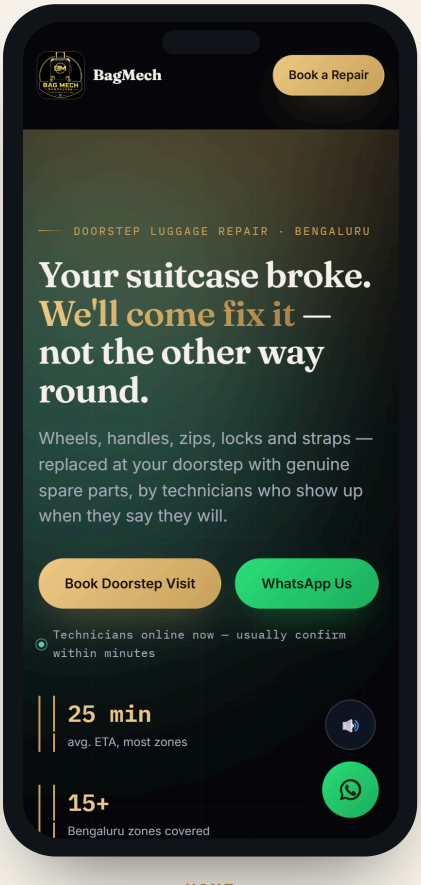


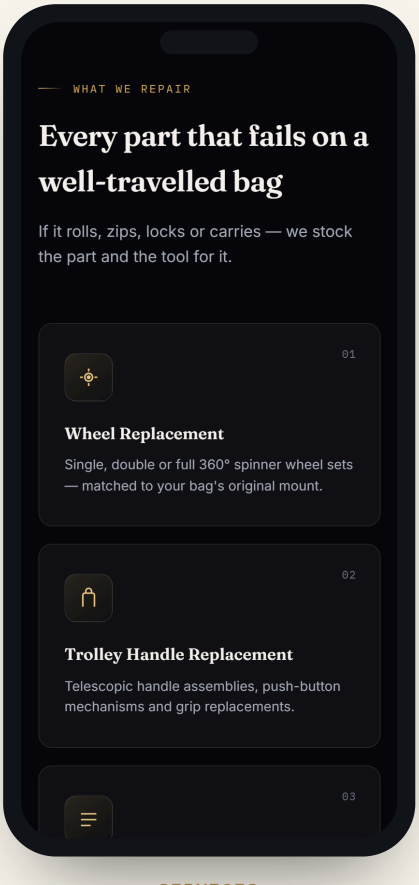
FIG 3.10 Footer with brand mark, quick links and a "Powered by CraftLanee" credit.

Designed for the phone in your hand

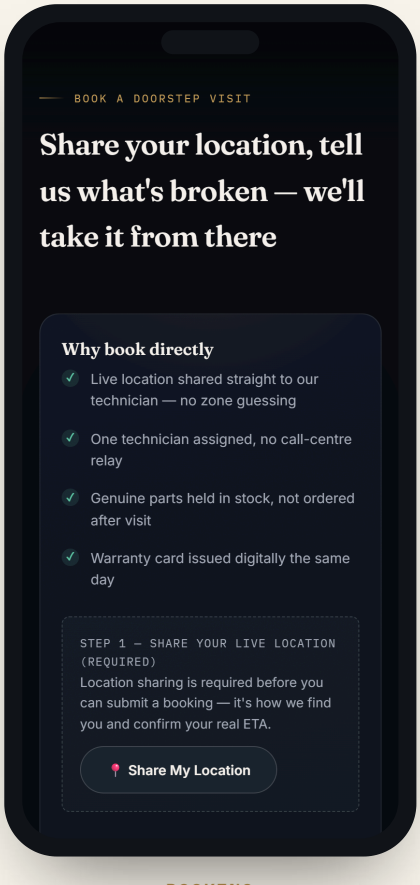
Most customers book from a phone, often while standing next to the broken bag. Every section reflows to a single column, buttons are thumb-sized, and the phone field opens a number keypad.



HOME



SERVICES



BOOKING

Touch-aware

Cursor and hover effects turn off on touch screens, so there are no wasted animations.

Motion-respectful

Visitors who set their device to "reduce motion" get a calm, static version of the site.

Lightweight

A single page with no front-end framework. Icons are inline SVG and sounds are generated in the browser, so there are no audio files to download.

From "my wheel broke" to the owner's phone in seconds

A booking is saved to the database first. Only after the customer has seen their confirmation are the alerts sent, in the background. A slow email service never delays the customer, and a failed alert never loses a booking.

1

Share location

Customer taps "Share My Location" and the browser returns GPS coordinates.

2

Fill the form

Name, phone, address, zone and repairs, checked in the browser.

3

Saved securely

The API validates the request and writes it to PostgreSQL as *pending*.

4

Confirmed

The customer sees a success message right away.

5

Owner alerted

Email (SendGrid) and WhatsApp (Twilio) are sent in the background.

WHAT THE OWNER RECEIVES ON WHATSAPP

New Booking Request — BagMech Bengaluru

Name: Ananya R.
Phone: 98765 43210
Zone: HSR Layout (~25 min ETA)
Location: 12.9116, 77.6389 (google.com/maps?q=...)
Doorstep Address: Flat 402, Sunrise Residency, 27th Main, HSR Layout
What needs repair: Wheel Replacement, Zip Replacement

Booking ID: 3f9c...e21a

10:42 ✓✓

Sample message with illustrative data. The same text goes out by email, with the repair and zone in the subject line.

Booking lifecycle

Each booking moves through five tracked statuses, and the API can update them. That is the groundwork for an owner dashboard.

`pending` `confirmed` `in_progress` `completed` `cancelled`

WhatsApp, the Indian default

Customers who prefer to chat can tap "WhatsApp Us". It opens a chat with the business and pre-fills a message with whatever they have already typed in the form: name, phone, address and repairs. The business number comes from server settings, so it can be changed without editing the site.

01

Exact pin, every time

Every web booking includes GPS coordinates and a Google Maps link, so the technician can navigate straight to it.

02

Owner stays on the road

New jobs reach the owner on WhatsApp while they are out on another repair. No dashboard to keep checking.

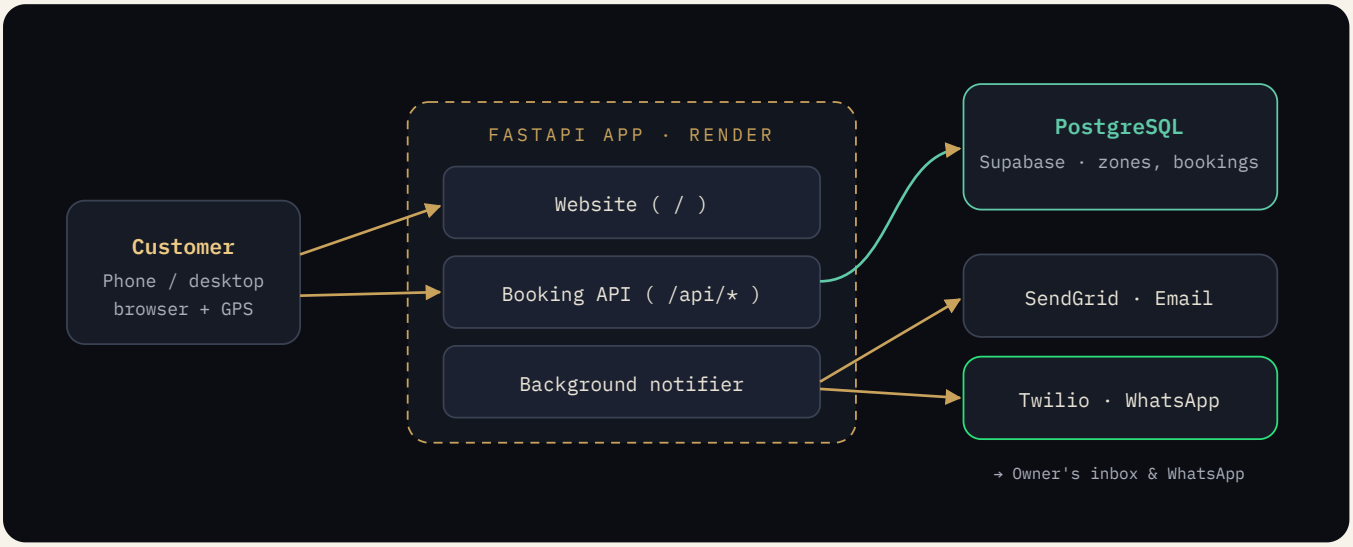
03

Nothing slips through

Every request is stored with a unique ID and a status, whether or not an alert gets through.

One small, fast service that is simple to run

A single Python application serves both the website and the booking API. One deployment and one URL keep hosting cheap and maintenance simple for a small business.



Stack

LAYER	TECHNOLOGY
Frontend	HTML5, modern CSS, vanilla JavaScript
Backend	Python 3.12, FastAPI, Uvicorn
Data	PostgreSQL on Supabase, async SQLAlchemy 2
Validation	Pydantic v2 schemas
Email	SendGrid HTTPS API
WhatsApp	Twilio WhatsApp API
Hosting	Render, with Git-based deploys

API endpoints

METHOD	PATH	PURPOSE
GET	/api/zones	Coverage zones + ETAs
POST	/api/bookings	Create a booking
GET	/api/bookings	List, filter by status
GET	/api/bookings/{id}	Single booking
PATCH	/api/bookings/{id}/status	Move through lifecycle
GET	/api/config	Public WhatsApp number
GET	/health	Uptime check

Built to keep running when things go wrong

Small businesses have no on-call engineer. The system is built so that common failures, such as a flaky email provider, a dropped database connection or planned downtime, are handled without anyone stepping in.

Data model

ZONES

`name`	Unique area name
`eta_minutes`	Average arrival time
`is_far`	Flags longer-ETA zones

BOOKINGS

`name, phone`	Customer contact
`zone_id`	Linked to zones
`address, issue`	Where and what
`latitude, longitude`	GPS pin, 6-decimal precision
`status`	Constrained to the 5 lifecycle values
`created_at`	Indexed for newest-first lists

Reliability features

- ✓ **Booking saved before alerts are sent.** A notification failure can never lose a customer.
- ✓ **Independent channels.** If email fails, WhatsApp still goes out, and the reverse.
- ✓ **Self-healing DB pool.** Connections are checked before use and recycled every 30 minutes.
- ✓ **Row-level security** on Supabase tables. Only the backend can write bookings.
- ✓ **Server-side validation** of every field, whatever the browser sends.
- ✓ **Offline fallback** for the zone list, so the page always renders.
- ✓ **All settings in environment variables.** No keys or phone numbers are hard-coded.

One-switch maintenance mode

Setting one environment variable pauses the whole site behind a clean "We'll be right back" page, with the correct HTTP 503 status. The health check keeps answering, so hosting monitors don't raise false alarms. The owner can pause bookings for a holiday or an outage without a developer.

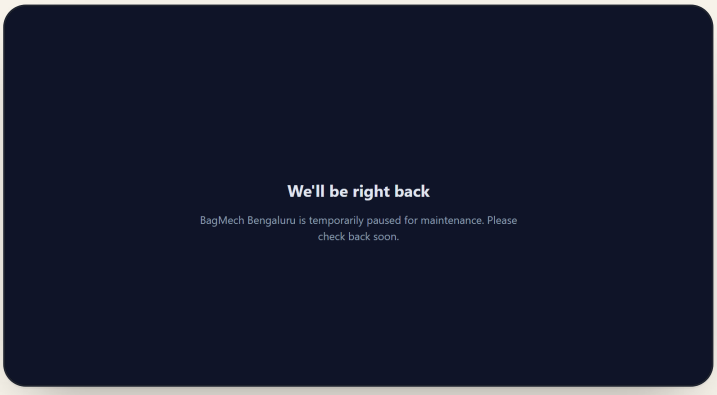


FIG 7.1 The maintenance page visitors see while the site is paused.

From first commit to live in six weeks

The work went in short, focused iterations. Each change was tested against the live hosting environment before moving on.

18 JUL 2026

Foundation

Full website and FastAPI backend, database schema and seeded coverage zones.

30 JUL 2026

Production hardening

Website served directly by the backend (one deployment). Async database driver fixes.

30 JUL 2026

Reliable email alerts

Moved from SMTP, which cloud hosts often block, to the SendGrid HTTPS API for dependable delivery.

31 JUL – 01 AUG

Brand finishing

BagMech logo, business phone and email, and the CraftLanee credit added.

22 AUG 2026

Maintenance mode

One-switch pause for the whole site, with a health check that stays up.

29 AUG 2026

Deployment pinning

Python version pinned for stable, repeatable deploys on Render.

30 AUG 2026

Live polish

Final asset and logo fixes on the production site.

HANDOVER CHECKLIST

What the client received

- ✓ Responsive brand website (desktop, tablet, mobile)
- ✓ GPS-first booking system
- ✓ 16-zone coverage with editable ETAs
- ✓ Email and WhatsApp booking alerts
- ✓ WhatsApp click-to-chat with pre-filled details
- ✓ Booking management API with a 5-stage lifecycle
- ✓ PostgreSQL schema with row-level security
- ✓ Maintenance mode and health check
- ✓ Cloud deployment and a run guide



DELIVERED · AUG 2026

Ready to grow into

The API already supports an owner dashboard: booking lists, status filters and status updates. Adding customer SMS updates, online payments or real technician tracking is a natural next phase.

DESIGNED & ENGINEERED BY

CraftLanee

CONNECT · COLLABORATE · CREATE

Have a service business that needs bookings, not just a website?

We design and build brand sites, booking flows and the backends behind them: fast, mobile-first, and connected to the channels your team already uses.

craftlanee.com